Seminar on Berkeley Lectures on p-adic Geometry

From classical deformation spaces to integral local Shimura varieties

This weekly, 17-session reading seminar follows Peter Scholze and Jared Weinstein's *Berkeley Lectures on p-adic Geometry* [SW20], from perfectoid spaces and diamonds to mixed-characteristic local shtukas and local Shimura varieties. Lubin–Tate, Drinfeld, and Rapoport–Zink spaces provide recurring examples. The goal is to understand their role in p-adic uniformization, the connection between tower cohomology and Langlands, and the passage to integral local and global Shimura varieties.

Lubin–Tate theory began with explicit local class field theory (1965) and formal-group deformations (1966). Drinfeld's spaces and Čerednik–Drinfeld uniformization supplied p-adic uniformizing domains; Rapoport–Zink's 1996 theory organized framed deformations and period maps. Following the infinite-level work of Faltings and Fargues, Scholze's perfectoid geometry and Scholze–Weinstein's work led to the local-shtuka framework. This geometry makes Frobenius, reduction, and deformation phenomena accessible within the arithmetic study of Shimura varieties [RZ96, RV14, SW13].

Format and preparation. Xiaowei Li gives the odd-numbered talks; Shenguanlin Lin gives the even-numbered talks. Prerequisites are schemes, local fields, and basic familiarity with reductive groups and étale cohomology. Vocabulary concerning p-divisible groups, Dieudonné modules, and quasi-isogenies will be recalled in Week 1 and before Week 8; Week 14 gives the fuller treatment. Talks emphasize constructions, representative proofs, and examples. Advanced foundations and recent integral-model theorems will be treated selectively.

Syllabus (tentative)

All lecture and section numbers below refer to the 2020 edition of [SW20]. Core readings specify the material to prepare; the talk should select its central constructions and representative proofs. Appendices and optional readings provide supplementary detail.

Week	Speaker	Content
1	Xiaowei Li	**Introduction and motivation.** Explain the route from Lubin–Tate and Drinfeld spaces to Rapoport–Zink spaces, local Shimura varieties, and local shtukas. Introduce the cohomological motivation from Langlands and the role of p-adic uniformization. Recall height, dimension, Tate modules, isocrystals, and quasi-isogenies. *Core reading:* [SW20], Lecture 1, §§1.1–1.2; [RV14], Introduction.
2	Shenguanlin Lin	**Adic spaces.** Huber pairs, continuous valuations, rational subsets, and the structure sheaf. Use open and closed discs to explain formal generic fibers. Treat analytic points and Cartier divisors as needed later; explain the sheafiness hypotheses used in constructing adic spaces. *Core reading:* [SW20], §§2.1–2.3, 3.1–3.3, 4.1–4.3, 5.1–5.3. *Optional:* [Bha17], §§7–8; the pre-adic-space appendix in [SW20].

Week	Speaker	Content (continued)
3	Xiaowei Li	**Perfectoid spaces.** Perfectoid rings and spaces, tilting, and examples of perfectoid fields and discs. Explain the tilting equivalence and invariance of the étale site; give the idea of almost purity and its use in towers. *Core reading:* [SW20], §§6.1–6.2 and 7.1–7.5. *Optional:* §6.3; [Bha17], §§3, 6, 9–10; [Sch12].
4	Shenguanlin Lin	**The Fargues–Fontaine curve.** Introduce the absolute curve attached to $F = C^\flat$, with $C/\mathbf{Q}_p$ complete and algebraically closed. Explain the interpretation of untilts as points, the rings A_{inf} and B^+_{dR}, vector bundles, and slopes. State the classification in this setting and illustrate modifications at an untilt point. Relative families return in Week 13. *Core reading:* [Ans], §§10–11 and 13 (constructions and main statements); [SW20], §13.5 (preview). *Optional:* [Ans], §§4–5, for A_{inf} and untilts.
5	Xiaowei Li	**Diamonds I.** Motivate diamonds as pro-étale quotients of perfectoid spaces. Define pro-étale morphisms and diamonds, and explain $\operatorname{Spd}\mathbf{Q}_p$ through untilts. Work through examples and the role of descent. *Core reading:* [SW20], §§8.1–8.4. *Optional:* [Sch17], definitions of diamonds and pro-étale morphisms.
6	Shenguanlin Lin	**Diamonds II.** Quasi-pro-étale morphisms, torsors, and the functor $X \mapsto X^\diamond$. Explain how rigid spaces are recovered in the diamond setting, and discuss underlying topological spaces and étale sites. *Core reading:* [SW20], §§9.1–9.4 and 10.1–10.4. *Optional:* The local-system appendix to Lecture 10; [Sch17].
7	Xiaowei Li	**Drinfeld shtukas over function fields.** Vector bundles with Frobenius modifications, legs, relative position, bounds, and level structures. Distinguish global shtukas on a curve from their local counterparts. Explain how partial Frobenius and cohomology enter the function-field Langlands story, without attempting its full proof. *Core reading:* [SW20], §1.1 and §11.1; revisit the examples of modifications from Week 4.
8	Shenguanlin Lin	**Mixed-characteristic shtukas.** Construct the geometric replacement for the product with $\operatorname{Spa}\mathbf{Z}_p$, and explain how an untilt supplies a leg. Define mixed-characteristic shtukas. Recall p-divisible groups over $\mathcal{O}_C$ and introduce the comparison with one-leg shtukas, integral Robba-ring modules, and de Rham data. *Core reading:* [SW20], Lectures 11–12, §§11.1–11.4 and 12.1–12.4.

Week	Speaker	Content (continued)
9	Xiaowei Li	**Shtukas with one leg.** Study the extension and classification results of Lectures 13–14. Relate one-leg shtukas to Breuil–Kisin–Fargues modules, and explain the minuscule conditions in the p-divisible-group comparison. Explain full faithfulness, essential surjectivity, and the role of extension across the closed point of $\operatorname{Spec} A_{\mathrm{inf}}$. *Core reading:* [SW20], §§13.1–13.5 and 14.1–14.4. *Optional:* The integral p-adic Hodge theory appendix, especially §14.8.
10	Shenguanlin Lin	**Examples of diamonds and Drinfeld's lemma.** The self-product of $\operatorname{Spd} \mathbf{Q}_p$, Banach–Colmez spaces, and their appearance in p-adic Hodge theory. Explain why a naive product formula for fundamental groups fails, and how Drinfeld's lemma repairs it through Frobenius. *Core reading:* [SW20], Lectures 15–16, §§15.1–15.2 and 16.1–16.3.
11	Xiaowei Li	**The v-topology and v-sheaves.** Define v-covers, small and spatial v-sheaves, and the relevant classes of morphisms. Explain the v-sheaves attached to perfect schemes and formal schemes, including the distinction between integral information and a generic-fiber diamond. *Core reading:* [SW20], Lectures 17–18, §§17.1–17.4 and 18.1–18.4. *Optional:* The Dieudonné-theory appendix to Lecture 17.
12	Shenguanlin Lin	**Affine Grassmannians and their families.** The B_{dR}^{+}-affine Grassmannian, Schubert varieties, and relative-position bounds. Explain the minuscule case, the role of Demazure resolutions, and convolution; introduce families over $\operatorname{Spd} \mathbf{Z}_p$ and products of leg spaces. *Core reading:* [SW20], Lectures 19–20, §§19.1–19.4 and 20.1–20.5. *Optional:* The G-torsor appendix to Lecture 19.
13	Xiaowei Li	**Affine flag varieties, local models, and relative bundles.** Parahoric affine flag varieties and the local-model construction; use a low-rank example to illustrate integral degeneration. Return to the relative Fargues–Fontaine curve: vector bundles, G-torsors, Newton semicontinuity, the étale locus, and extension results. *Core reading:* [SW20], Lectures 21–22, §§21.1–21.5 and 22.1–22.6. *Optional:* Lecture 21's examples; [PRS13], introductory examples.

Week	Speaker	Content (continued)
14	Shenguanlin Lin	**p-divisible groups and Rapoport–Zink spaces.** Review Dieudonné theory, deformation theory, and framed EL/PEL moduli problems. Compare the height-zero Lubin–Tate open ball and Drinfeld half-space at their natural levels over $\breve{\mathbf{Q}}_p$. Distinguish crystalline and Hodge–Tate period maps, admissibility, and weak admissibility. Explain duality of the full towers at infinite level over $\mathbf{C}_p$. *Core reading:* [SW20], §§24.2–24.3 (classical moduli side); [SW13], §§6.1, 6.4 and 7.1–7.2, especially Theorem 7.2.3. *Optional:* [RZ96], Chapter 3; [RV14], §§3–5.
15	Xiaowei Li	**Introduction to global Shimura varieties.** Shimura data, complex uniformization, canonical models, and integral models. Explain Serre–Tate deformation theory and basic Rapoport–Zink uniformization. Use a modular or quaternionic Shimura curve as an example; distinguish basic tubes from the whole global variety and indicate the role of Igusa varieties in nonbasic uniformization. *Core reading:* [Mil], §§5–6 and 12 (definitions and examples); [RV14], Remark 5.9; [RZ96], Chapter 6 (main statements). *Optional:* [PR24], Introduction, for the integral local/global bridge.
16	Shenguanlin Lin	**Local Shimura varieties.** Construct moduli spaces of bounded local shtukas, emphasizing one leg and minuscule bounds. Introduce $(G, b, \{\mu\})$, $B(G, \mu)$, the reflex field E, base $\breve{E}$, level K, and the actions of $G(\mathbf{Q}_p)$ and $J_b(\mathbf{Q}_p)$ on the tower. Explain the EL/PEL comparison and period maps with their levels and conventions. *Core reading:* [SW20], Lectures 23–24, especially §§23.1–23.3 and 24.1–24.3, including Corollary 24.3.5; survey §§23.4–23.5.
17	Xiaowei Li	**Integral models of local Shimura varieties.** Integral local shtukas, parahoric data, and the EL/PEL comparison. Explain how the local model depends on $(\mathcal{G}, \mu)$ while the integral local Shimura variety also depends on b. Discuss representability and local singularities. Conclude with selected developments since 2020 and their relevance to global integral models. *Core reading:* [SW20], Lecture 25, §§25.1–25.5; [PR26], §2.5, especially the results 2.5.4–2.5.7 with their hypotheses. *Optional:* [AGLR], Introduction; [PR24], local/global comparison statements.

Expected outcomes

Participants should be able to explain the EL/PEL generic-fiber comparison with its level [SW20, Corollary 24.3.5], distinguish the period maps, and relate local moduli to basic uniformization. The height-two Lubin–Tate space and the GL_2 Iwahori local model provide concrete examples. Lecture 25 will be compared with [PR26]: its representability theorems treat abelian-type data at odd p and a specified type A and C range at $p = 2$, with quasi-parahoric level and the precise hypotheses of §2.5. Advanced proofs remain further reading.

References

Main text and core companions. [SW20] is the main text. The other entries in this group are used only in the selected portions specified above.

[**SW20**] P. Scholze and J. Weinstein, *Berkeley Lectures on p-adic Geometry*, Annals of Mathematics Studies 207, Princeton University Press, 2020. Authors' PDF.

[**SW13**] P. Scholze and J. Weinstein, *Moduli of p-divisible groups*, Cambridge Journal of Mathematics **1** (2013), 145–237. arXiv:1211.6357.

[**RV14**] M. Rapoport and E. Viehmann, *Towards a theory of local Shimura varieties*, Münster Journal of Mathematics **7** (2014), 273–326. arXiv:1401.2849.

[**RZ96**] M. Rapoport and T. Zink, *Period Spaces for p-divisible Groups*, Annals of Mathematics Studies 141, Princeton University Press, 1996.

Supporting readings. These supply background and examples; they are not additional texts to read in full.

[**Bha17**] B. Bhatt, *Lecture notes for a class on perfectoid spaces*, course notes, 2017. Lecture notes.

[**Mil**] J. S. Milne, *Introduction to Shimura Varieties*, lecture notes, revised 2017. Author's PDF.

[**Ans**] J. Anschütz, *Lectures on the Fargues–Fontaine curve*, lecture notes. Lecture notes.

[**Sch12**] P. Scholze, *Perfectoid spaces*, Publications Mathématiques de l'IHÉS **116** (2012), 245–313. arXiv:1111.4914.

[**Sch17**] P. Scholze, *Étale cohomology of diamonds*, arXiv:1709.07343. A reference for the foundations, not mandatory cover-to-cover reading.

[**PRS13**] G. Pappas, M. Rapoport, and B. Smithling, *Local models of Shimura varieties, I. Geometry and combinatorics*, in *Handbook of Moduli*, Vol. III, 2013, 135–217. arXiv:1011.5551.

Integral-model endpoint and optional outlook. Read the introductions and selected main statements. Detailed proofs lie beyond the scope of the final session.

[**PR24**] G. Pappas and M. Rapoport, *p-adic shtukas and the theory of global and local Shimura varieties*, Cambridge Journal of Mathematics **12** (2024), 1–164. arXiv:2106.08270.

[**AGLR**] J. Anschütz, I. Gleason, J. Lourenço, and T. Richarz, *On the p-adic theory of local models*, arXiv:2201.01234 (2022; revised 2026).

[**PR26**] G. Pappas and M. Rapoport, *On integral local Shimura varieties*, Journal of the Institute of Mathematics of Jussieu **25** (2026), 375–443. arXiv:2204.02829 (final preprint version, 2025).